

# Literature status: the measurable-stability formulation asked for in arXiv:1305.1819

Answer identified in arXiv:1804.09099

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**Status: already answered in later literature.** The future-work request in arXiv:1305.1819 is answered by DeCorte, de Oliveira Filho, and Vallentin, arXiv:1804.09099. Their completely-positive convex program has optimal value exactly equal to the measurable independence number; they identify its dual cone as the cone of copositive kernels and explicitly apply the theorem to the sphere orthogonality graph.

## 1 Question in the source

The final section, “Future work,” of Dobre, Dür, Frerick, and Vallentin [1] asks for a copositive formulation of the measurable stability problem for graphs admitting infinite measurable stable sets. Its central example is the graph on the unit sphere in which two vertices are adjacent exactly when they are orthogonal. Thus the requested parameter is

$$m_0(S^{n-1}) = \sup\{\omega(I) : I \subseteq S^{n-1} \text{ measurable and containing no orthogonal pair}\},$$

where  $\omega$  is surface measure. This is Witsenhausen’s problem. The source does not ask for the still-open numerical evaluation of  $m_0(S^{n-1})$ ; it asks for an exact copositive optimization formulation from which stronger bounds could be derived.

## 2 Exact later answer

DeCorte, de Oliveira Filho, and Vallentin [2] address precisely this setting. The match is explicit at four levels.

1. Section 2, example (ii), identifies Witsenhausen’s problem with the measurable independence number  $\alpha_\omega(G(S^{n-1}, \{\pi/2\}))$ .
2. Their general convex program (equation (3.3) in the published numbering; the arXiv source label is **theta-problem**) maximizes

$$\langle J, A \rangle$$

over continuous kernels  $A$  with unit diagonal integral, vanishing on graph edges, and lying in a chosen cone. For the cone  $\mathcal{C}(V)$  of completely positive kernels, it is the infinite measurable analogue of the standard finite completely-positive stability-number program.

3. Section 4 defines the dual cone  $\mathcal{C}^*(V)$  and explicitly proves that it is the cone of copositive kernels:

$$\mathcal{C}^*(V) = \{Z : \langle Z, f \otimes f^* \rangle \geq 0 \text{ for every } f \in L^2(V), f \geq 0\}.$$

Thus the program has exactly the copositive/completely-positive conic duality requested in the 2013 paper.

4. Theorem 5.1 (labelled `thm:cp-exactness`) proves

$$\vartheta(G, \mathcal{C}(V)) = \alpha_\omega(G)$$

for locally independent compact vertex-transitive graphs under the stated Haar measure and density-metric hypotheses. The paragraph immediately following the theorem applies it to every single-angle graph  $G(S^{n-1}, \{\theta\})$ . Taking  $\theta = \pi/2$  is exactly the sphere orthogonality graph.

The paper also states the specialization directly as Theorem 1.1: replacing positive type by completely positive type in its spherical optimization problem makes the optimal value exactly  $m_0(S^{n-1})$ . In radial form this is

$$\begin{aligned} & \text{maximize} && \int_{S^{n-1}} \int_{S^{n-1}} f(x \cdot y) d\omega(y) d\omega(x) \\ & \text{subject to} && f(1) = \omega(S^{n-1})^{-1}, \quad f(0) = 0, \\ & && f \text{ continuous and of completely positive type for } S^{n-1}. \end{aligned}$$

This is an exact formulation, not merely an upper bound. The numerical value of Witsenhausen’s parameter (and the double-cap conjecture) remains open; that does not weaken the answer to the formulation request.

### 3 Conclusion and verification

The 2018 arXiv preprint, later published in *Mathematical Programming*, therefore resolves the future-work question in arXiv:1305.1819 in the intended sense. It is especially strong evidence of continuity of purpose that Frank Vallentin is a coauthor of both papers and that the later paper cites the 2013 copositive-kernel work while extending it to measurable independence.

The identification was checked against the official PDFs and TeX sources of both papers, including the 2013 “Future work” section; the 2018 Section 2 Witsenhausen example; the general program; Section 4’s copositive dual cone; Theorem 5.1 and its immediate spherical application; and Theorem 1.1.

### References

- [1] C. Dobre, M. Dür, L. Frerick, and F. Vallentin, *A copositive formulation for the stability number of infinite graphs*, arXiv:1305.1819, 2013.
- [2] E. DeCorte, F. M. de Oliveira Filho, and F. Vallentin, *Complete positivity and distance-avoiding sets*, arXiv:1804.09099; *Mathematical Programming* 191 (2022), 487–558.